Executive Overview – E-commerce Return Performance

100K	14K	14.05%	$382.69M
Total Orders	Returned Orders	Return Rate	Sum of Order_Value

Year Month
- 2024-01
- 2024-02

Region
- Central
- East

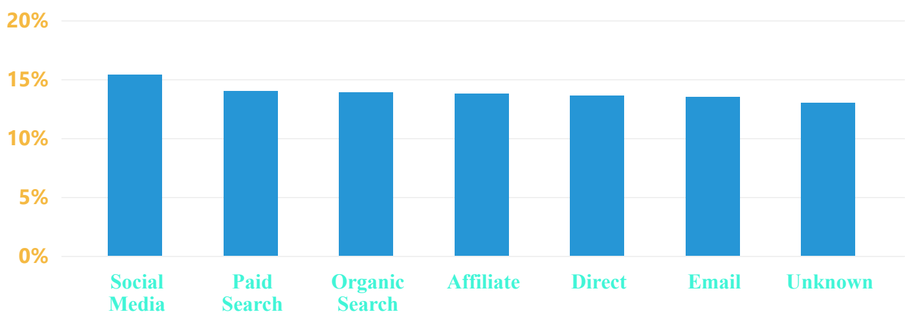
State
- Andhra Pradesh
- Bihar
- Chhattisgarh

Category
- Beauty
- Books

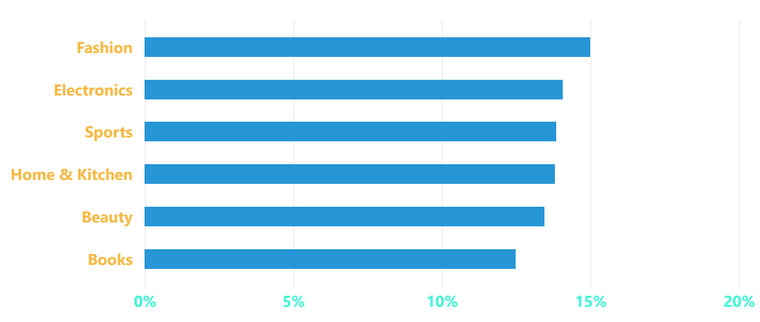
Payment_Method
- Cash on Delivery
- Credit Card

Shipping_Method
- Express
- Same Day
- Standard

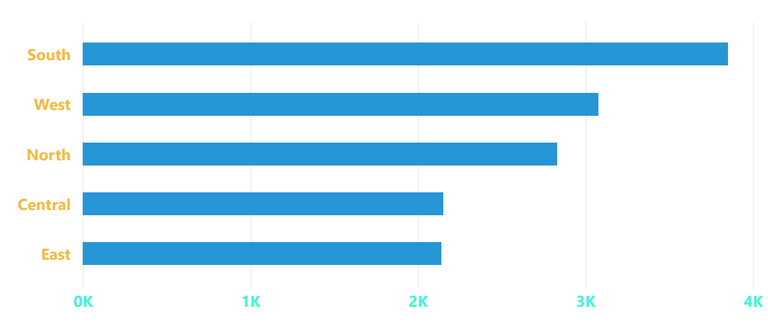
Return Rate, Total Orders and Returned Orders by Marketing_Channel



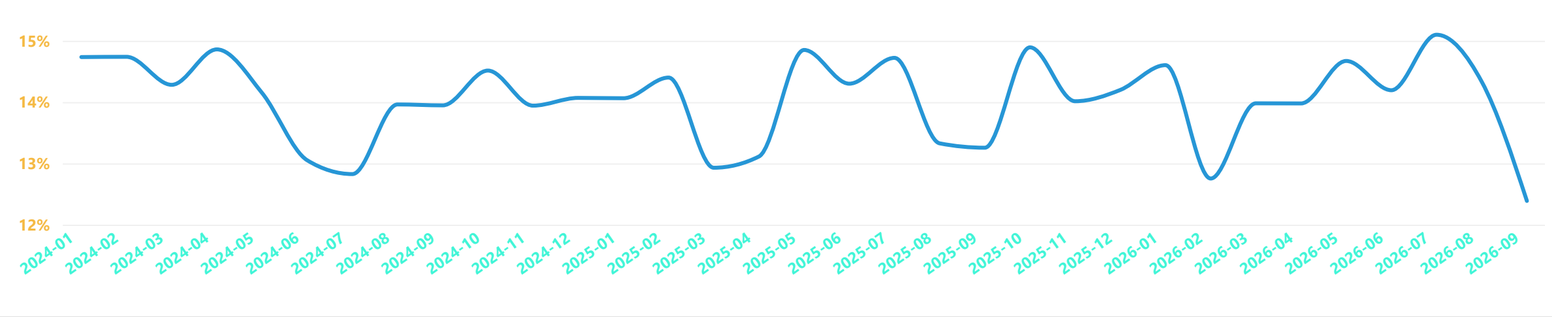
Return Rate by Category



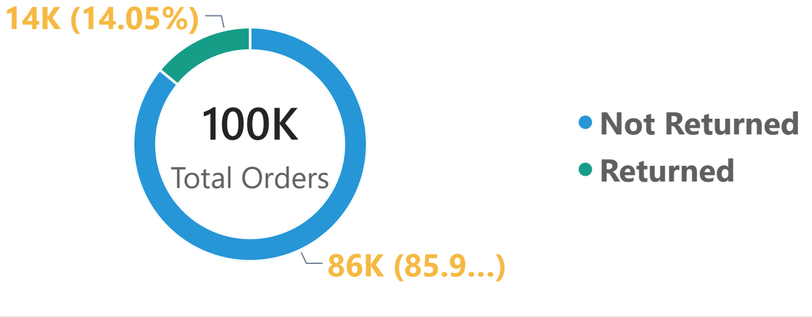
Returned Orders, Total Orders and Return Rate by Region



Return Rate by Year Month

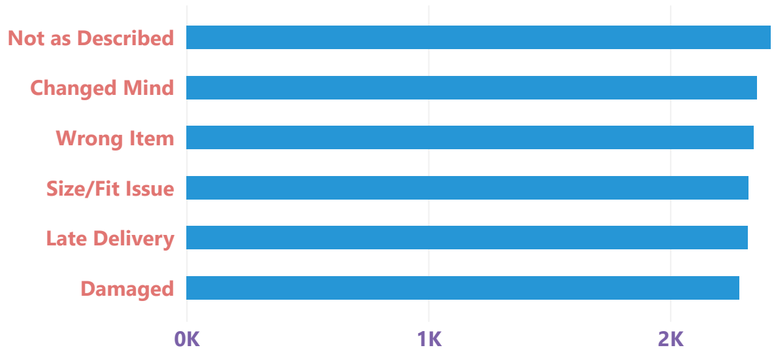


Total Orders, Total Orders and Return Rate by Return_Status



Customer Experience & Fulfilment Analysis

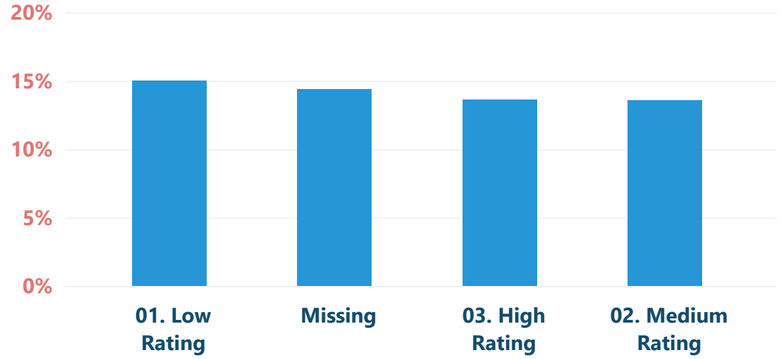
Return Reason Orders and Return Reason Share by Return_Reason



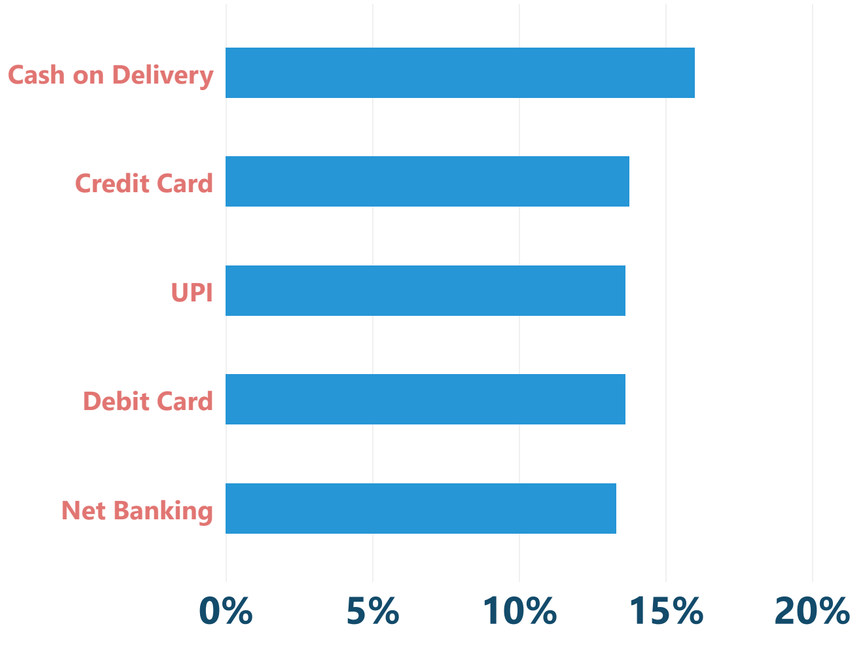
Return Rate by Delivery Group



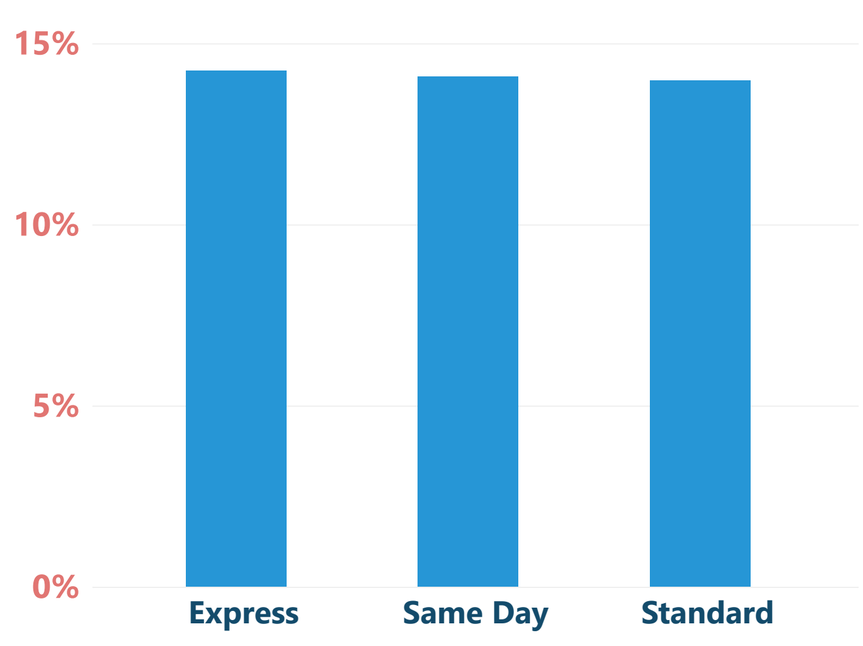
Return Rate, Returned Orders and Total Orders by Rating Group



Return Rate and Total Orders by Payment_Method



Return Rate and Total Orders by Shipping_Method

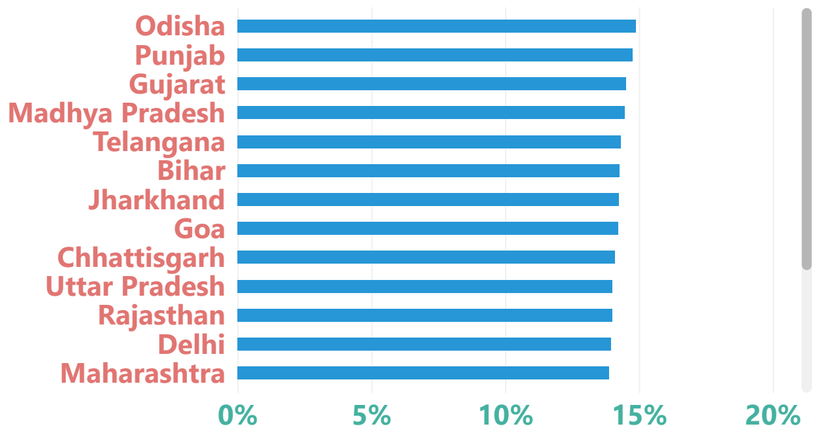


Return Rate and Total Orders by Quantity

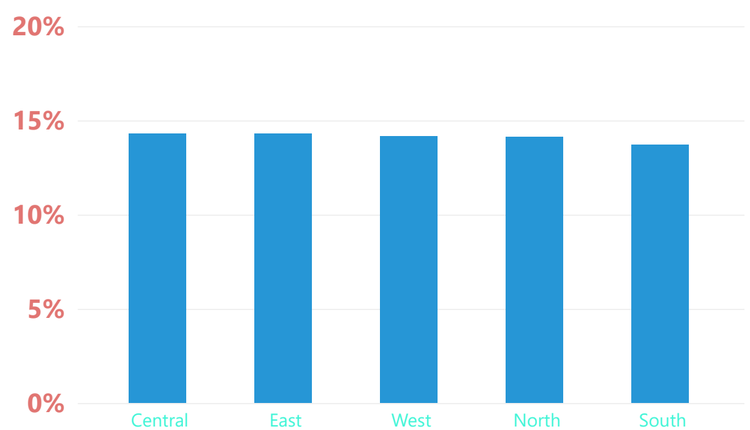


Geographic Return Performance & Market Insights

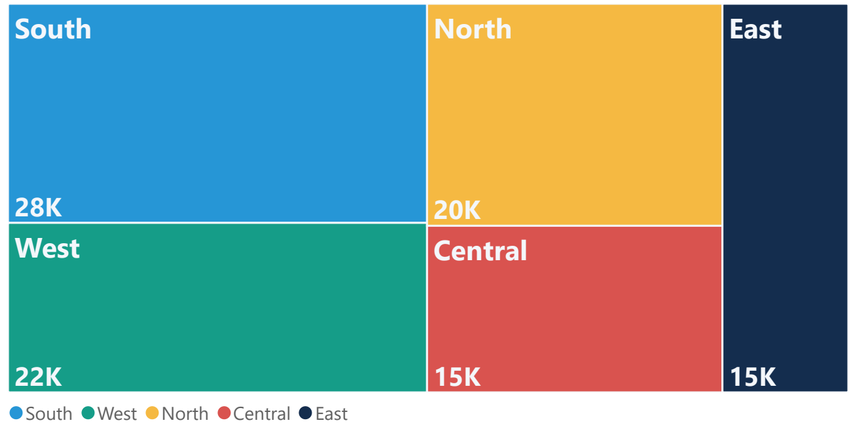
Return Rate, Total Orders and Returned Orders by State



Return Rate, Returned Orders and Total Orders by Region



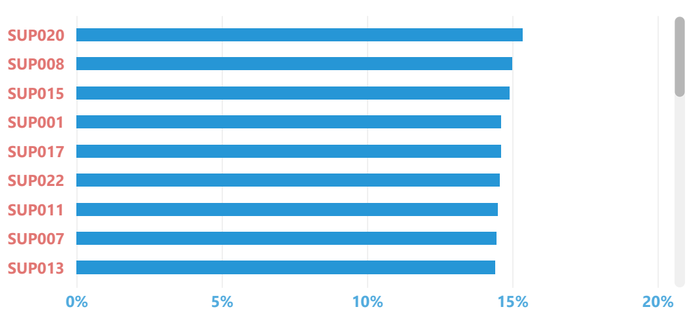
Total Orders, Return Rate and Returned Orders by Region



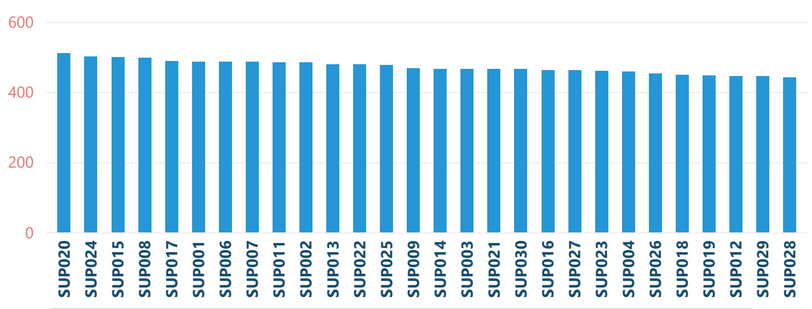
State	Andhra Pradesh				Bihar				Chhattisgarh				Delhi				Goa	
Region	Total Orders	Returned Orders	Return Rate	Total Order Value	Total Orders	Returned Orders	Return Rate	Total Order Value	Total Orders	Returned Orders	Return Rate	Total Order Value	Total Orders	Returned Orders	Return Rate	Total Order Value	Total Orders	Returned
Central									7611	1073	14.10%	2,92,64,812.33						
East					3769	538	14.27%	1,39,67,006.09										
North													4945	690	13.95%	1,88,90,031.32		
South	5644	766	13.57%	2,15,62,957.03														
West																	5453	
Total	**5644**	**766**	**13.57%**	**2,15,62,957.03**	**3769**	**538**	**14.27%**	**1,39,67,006.09**	**7611**	**1073**	**14.10%**	**2,92,64,812.33**	**4945**	**690**	**13.95%**	**1,88,90,031.32**	**5453**	

Supplier & Product Quality Intelligence Supplier performance, category and product analysis

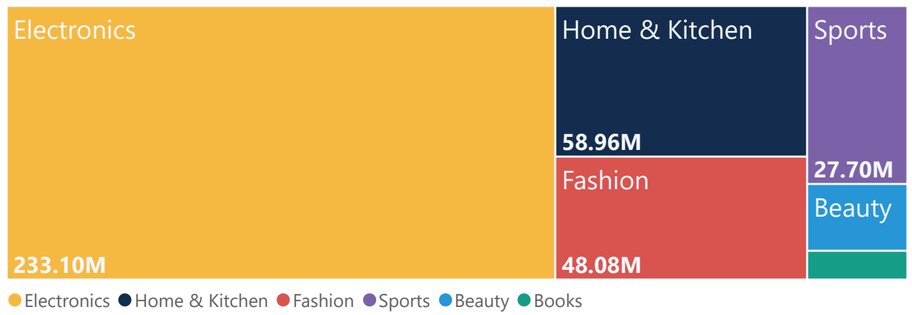
Return Rate, Total Orders and Returned Orders by Supplier_ID



Returned Orders, Return Rate and Total Orders by Supplier_ID

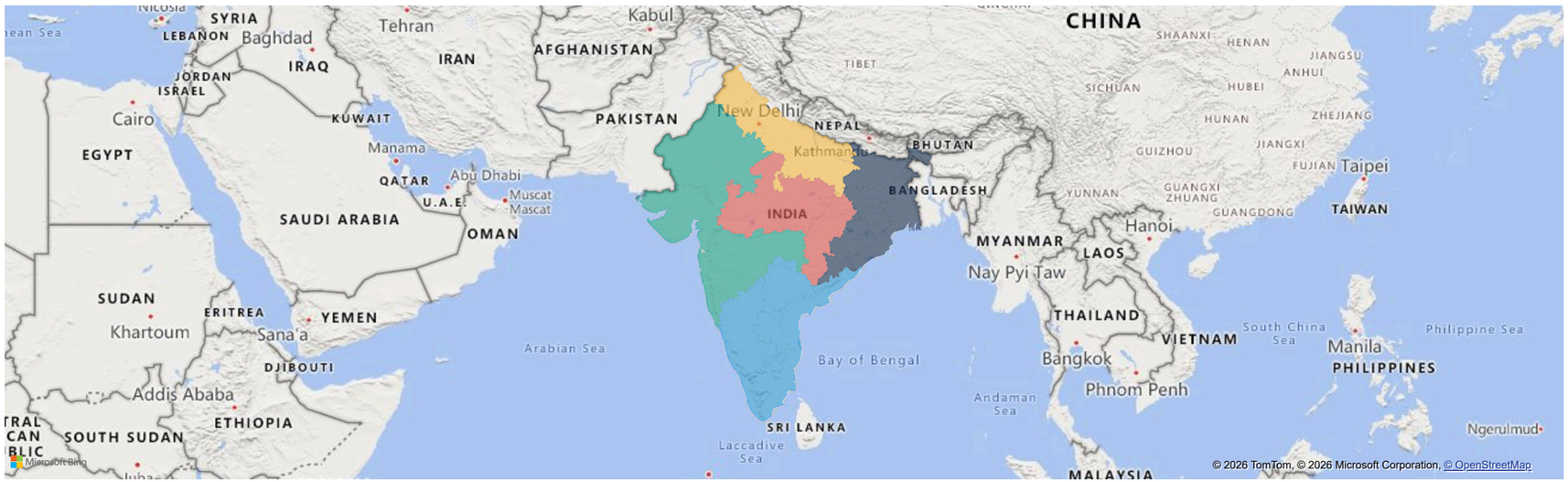


Total Order Value, Return Rate and Total Orders by Category



Product_ID	Category	Supplier_ID	Total Orders	Returned Orders	Return Rat
PRD00001	Beauty	SUP001	8	1	12.50'
PRD00001	Books	SUP001	4	1	25.00'
PRD00001	Electronics	SUP001	7		
PRD00001	Fashion	SUP001	12	1	8.33'
PRD00001	Home & Kitchen	SUP001	11	1	9.09'
PRD00001	Sports	SUP001	4	2	50.00'
PRD00002	Beauty	SUP002	4	1	25.00'
PRD00002	Books	SUP002	2		
PRD00002	Electronics	SUP002	10	2	20.00'
PRD00002	Fashion	SUP002	11	5	45.45'
PRD00002	Home & Kitchen	SUP002	8		
PRD00002	Sports	SUP002	1		
PRD00003	Beauty	SUP003	6	1	16.67'
PRD00003	Books	SUP003	2	1	50.00'
PRD00003	Electronics	SUP003	7	1	14.29'
Total			**100000**	**14047**	**14.05'**

Return Rate, Returned Orders and Total Orders by State and Region

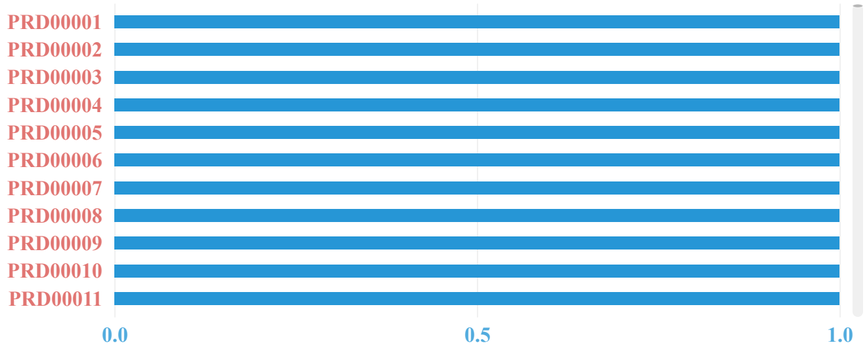


Central East North South West

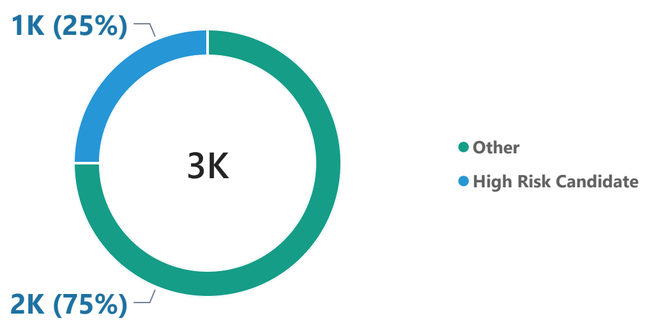
Predictive Return Risk & Model Evaluation Logistic Regression scores, high-risk products, model metrics

Risk Products Scored	High Risk Products	Orders Scored	Average Predicted Return Risk
3K	625	20K	49.67%

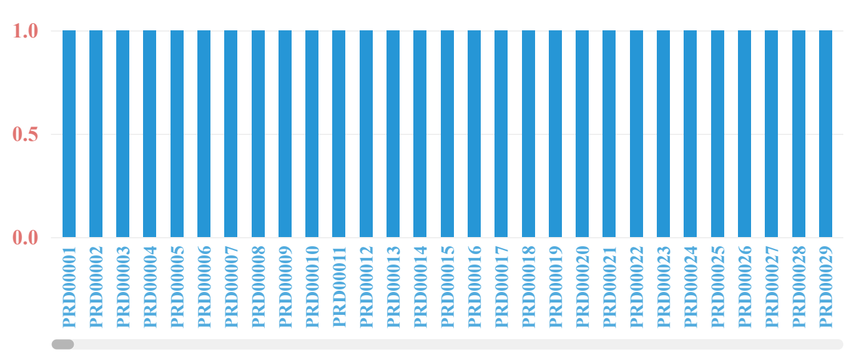
Count of Average_Return_Risk, Orders Scored, Count of Actual_Returns and Count of Maximum_Return_Risk by Product_ID



Distinct Products by Risk_Level

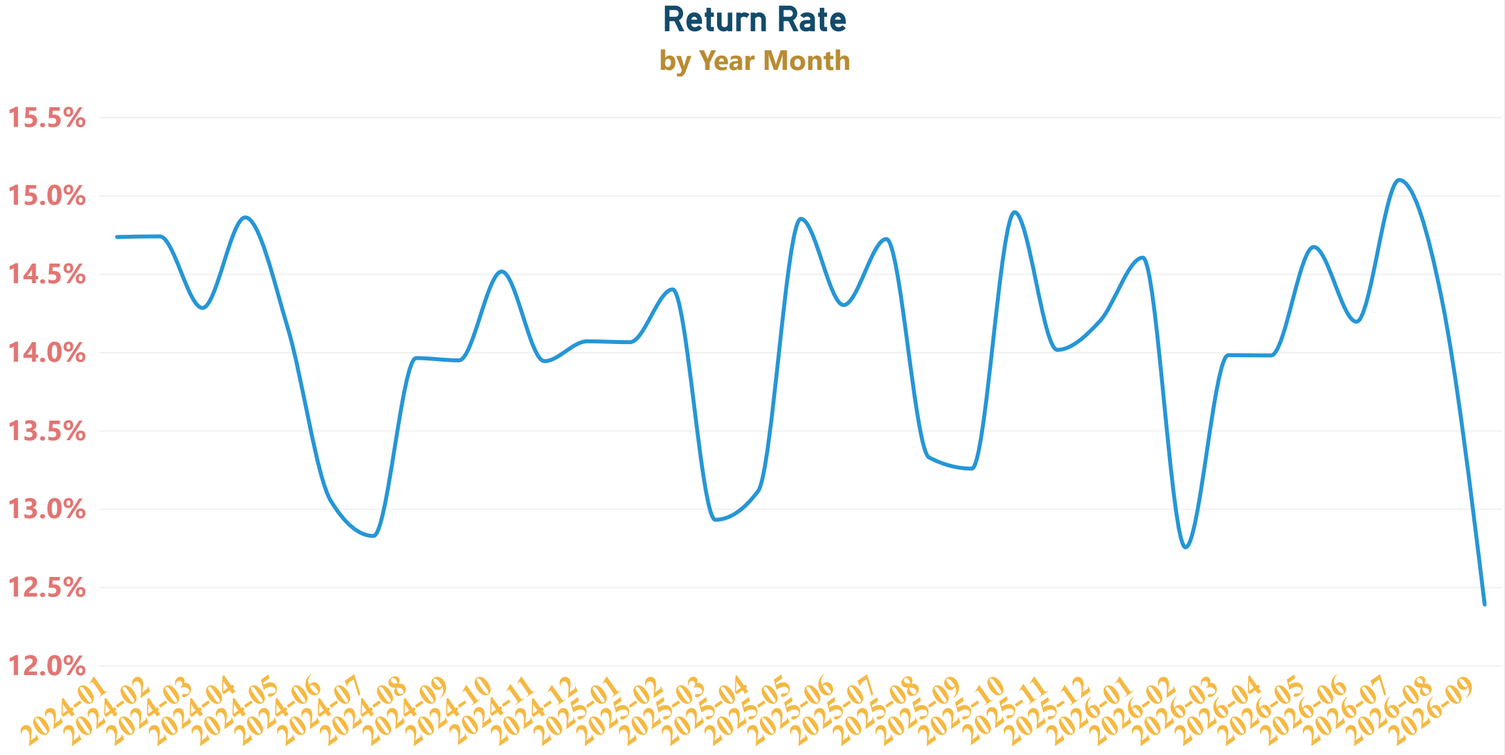


Count of Average_Return_Risk, Count of Orders_Scored and Count of Actual_Returns by Product_ID



Product_ID	Orders_Scored	Actual_Returns	Average_Return_Risk	Maximum_Return_Risk	Risk_Level
PRD00001	8	2	0.49	0.53	Other
PRD00002	5	3	0.53	0.58	High Risk Candidate
PRD00003	11	2	0.47	0.53	Other
PRD00004	5	1	0.48	0.54	Other
PRD00005	7	1	0.52	0.58	High Risk Candidate
PRD00006	3	1	0.51	0.57	High Risk Candidate
PRD00007	7	0	0.51	0.60	High Risk Candidate
PRD00008	14	3	0.51	0.60	High Risk Candidate
PRD00009	12	1	0.49	0.56	Other
PRD00010	5	0	0.48	0.50	Other

Product-Level Return Investigation Individual product





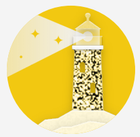
Order_ID	Order_Date	Sum of Quantity	Sum of Order_Value	Sum of Is_Returned
ORD0000001	16 September 2024	1	$507.62	0
ORD0000002	13 February 2025	1	$2,076.14	0
ORD0000003	07 October 2024	3	$1,748.17	0
ORD0000004	24 April 2025	1	$381.88	0
ORD0000005	25 June 2026	2	$19,070.03	0
ORD0000006	16 December 2025	1	$283.20	1
ORD0000007	31 August 2025	1	$554.09	0
ORD0000008	11 October 2024	1	$1,063.96	0
Total		**137942**	**$38,26,90,138.28**	**14047**

Return Drivers & Category Trend Analysis

Returned Orders
14047

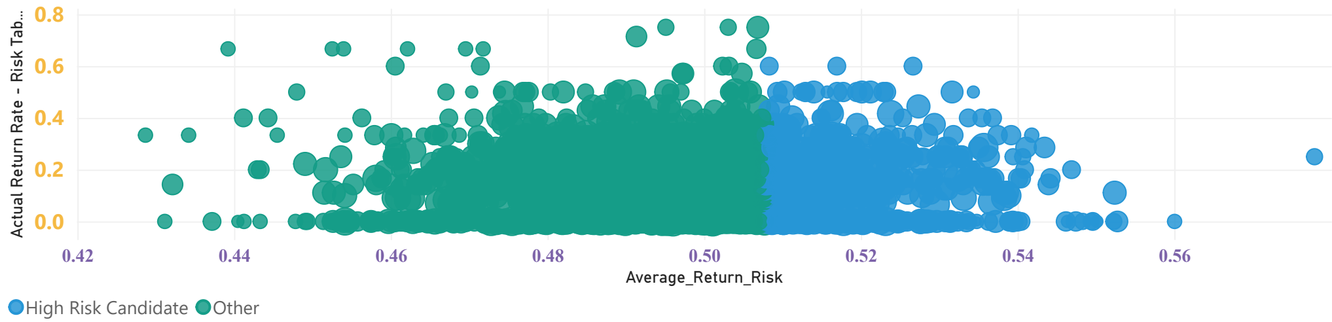
Key influencers | Top segments

What influences Is_Returned to [Increase] ?



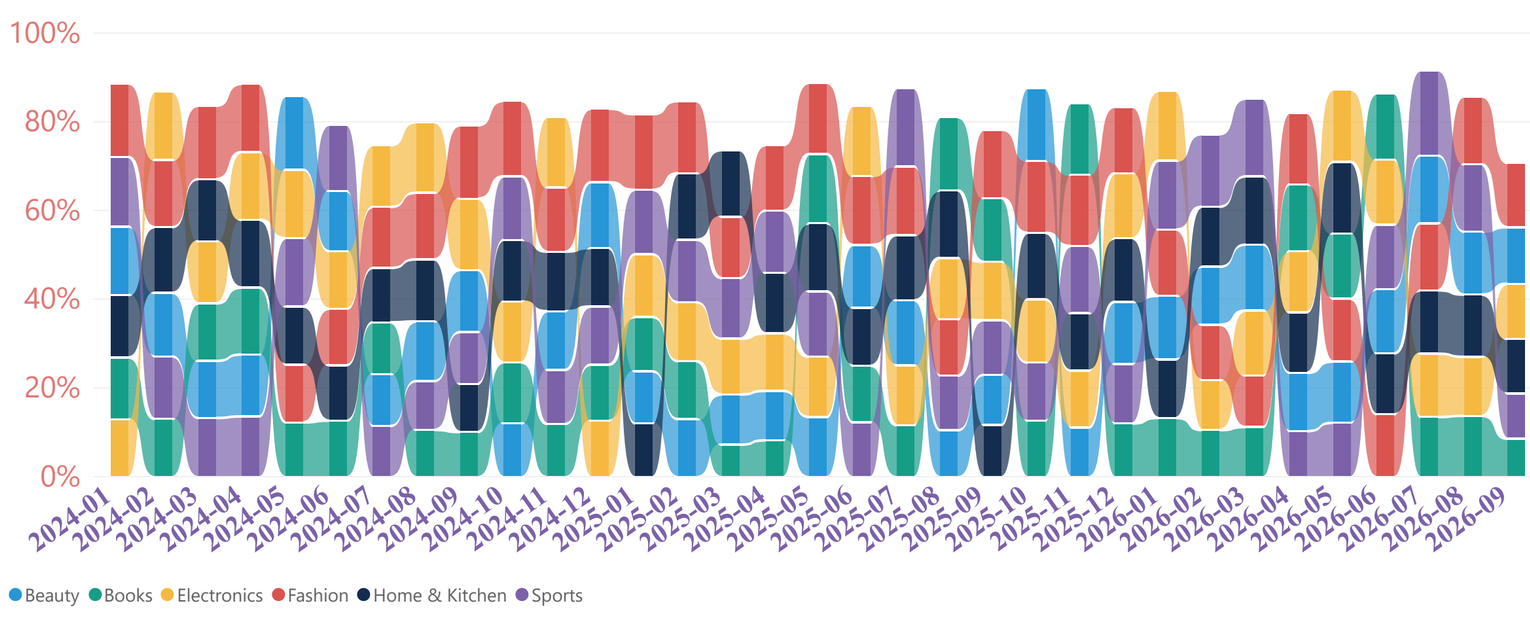
No influencers found. Try adding some more fields into 'Explain by'.

Actual Return Rate - Risk Table, Orders Scored, Count of Maximum_Return_Risk and Count of Actual_Returns
by Risk_Level and Average_Return_Risk



Category	2024-01	2024-02	2024-03	2024-04	2024-05	2024-06	2024-07	202
Beauty	15.45%	14.33%	12.86%	13.95%	16.62%	13.57%	11.68%	1
Books	13.96%	12.56%	12.96%	15.09%	11.79%	12.17%	11.69%	1
Electronics	12.46%	15.42%	13.99%	15.28%	15.41%	13.00%	13.90%	1
Fashion	16.57%	15.16%	16.55%	15.36%	13.00%	12.69%	13.75%	1
Home & Kitchen	14.07%	14.81%	14.02%	15.27%	13.13%	12.52%	12.25%	1
Sports	15.67%	14.09%	12.78%	13.12%	15.41%	14.94%	10.98%	1
Total	**14.73%**	**14.74%**	**14.28%**	**14.86%**	**14.15%**	**13.05%**	**12.83%**	**1**

Return Rate, Returned Orders and Total Orders
by Year Month and Category



Business insights and project report

Project objective

To identify key factors associated with e-commerce product returns, compare return performance across categories, suppliers, regions and marketing channels, and use Logistic Regression to prioritise products with elevated predicted return risk.

Key findings from verified SQL analysis

Finding Result

Overall return rate 14.05%

Highest category return rate Fashion — 14.99%

Highest channel return rate Social Media — 15.34%

Highest delivery-group rate Over 6 days — 16.91%

Lowest delivery-group rate Up to 3 days — 11.88%

Low customer rating return rate 15.00%

ML findings

2,500 products scored.

625 products identified as top-quartile high-risk candidates.

Test PR-AUC 0.167733.

The model provides modest lift over baseline and requires further improvement before operational deployment.

Business recommendations

Review product descriptions, size and fit information in the Fashion category.

Investigate social-media campaign traffic and product expectations.

Monitor products with delayed delivery.

Review high-risk products with their actual order counts before taking supplier action.